Nombre: ___ Elías Niebla López ___ Matrícula:___ A01641819 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.6000 < X < 50.2000) = P(-1.2807 < Z < 2.2443) = 0.8875$

3. $X \sim Bin(n = 14, p = 0.05)$
   $P(X = 3) = 0.025880$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.6000-49.4000)/(6*0.1702) = 1.1750
   Cpks = (50.6000-49.8180)/(3*0.1702) = 1.5314
   Cpki = (49.8180-49.4000)/(3*0.1702) = 0.8186
   Cpk = min(1.5314, 0.8186) = 0.8186

5. **** Diferencia de medias
   $\nu = 68.3055$
   LCS = 3.60 - 2.5900 + 1.6676(0.3087) = 1.0100 + 0.5148 = 1.5248
   LCI = 3.60 - 2.5900 - 1.6676(0.3087) = 1.0100 - 0.5148 = 0.4952
   ** Antes del cambio
   LCS = 3.6000 + 1.6849(1.0900/6.3246**0.5) = 3.6000 + 0.2904 = 3.8904
   LCI = 3.6000 - 1.6849(1.0900/6.3246**0.5) = 3.6000 - 0.2904 = 3.3096

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0930$
   $\mu = 500 - 8(-1.3225) = 510.5800$

Nombre: ___ Diego Iván Martínez Camero ___ Matrícula:___ A01742276 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.9000 < X < 50.4000) = P(0.4818 < Z < 3.4193) = 0.3147$

3. $X \sim Bin(n = 15, p = 0.05)$
   $P(X = 4) = 0.004853$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.6000-49.4000)/(6*0.1702) = 1.1750
   Cpks = (50.6000-49.8180)/(3*0.1702) = 1.5314
   Cpki = (49.8180-49.4000)/(3*0.1702) = 0.8186
   Cpk = min(1.5314, 0.8186) = 0.8186

5. **** Diferencia de medias
   $\nu = 68.3055$
   LCS = 3.60 - 3.1400 + 1.9127(0.3087) = 0.4600 + 0.5905 = 1.0505
   LCI = 3.60 - 3.1400 - 1.9127(0.3087) = 0.4600 - 0.5905 = -0.1305
   ** Antes del cambio
   LCS = 3.6000 + 1.9371(1.0900/6.3246**0.5) = 3.6000 + 0.3338 = 3.9338
   LCI = 3.6000 - 1.9371(1.0900/6.3246**0.5) = 3.6000 - 0.3338 = 3.2662

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0460$
   $\mu = 500 - 8(-1.6849) = 513.4795$

Nombre: ___ Diego Coronel Hinojosa ___ Matrícula:___ A01743668 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.8000 < X < 50.3000) = P(-0.1058 < Z < 2.8317) = 0.5398$

3. $X \sim Bin(n = 15, p = 0.05)$
   $P(X = 7) = 0.000003$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.6000-49.4000)/(6*0.1702) = 1.1750
   Cpks = (50.6000-49.8180)/(3*0.1702) = 1.5314
   Cpki = (49.8180-49.4000)/(3*0.1702) = 0.8186
   Cpk = min(1.5314, 0.8186) = 0.8186

5. **** Diferencia de medias
   $\nu = 70.0569$
   LCS = 3.60 - 2.7600 + 1.8401(0.3049) = 0.8400 + 0.5611 = 1.4011
   LCI = 3.60 - 2.7600 - 1.8401(0.3049) = 0.8400 - 0.5611 = 0.2789
   ** Antes del cambio
   LCS = 3.6000 + 1.8617(1.0900/6.4031**0.5) = 3.6000 + 0.3169 = 3.9169
   LCI = 3.6000 - 1.8617(1.0900/6.4031**0.5) = 3.6000 - 0.3169 = 3.2831

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1110$
   $\mu = 500 - 8(-1.2212) = 509.7698$

Nombre: ___ María José Cárdenas Acosta ___ Matrícula:___ A01742297 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(50.0000 < X < 50.1000) = P(1.0693 < Z < 1.6568) = 0.0937$

3. $X \sim Bin(n = 15, p = 0.05)$
   $P(X = 4) = 0.004853$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227

5. **** Diferencia de medias
   $\nu = 82.3169$
   LCS = 3.60 - 2.6800 + 1.9072(0.2818) = 0.9200 + 0.5375 = 1.4575
   LCI = 3.60 - 2.6800 - 1.9072(0.2818) = 0.9200 - 0.5375 = 0.3825
   ** Antes del cambio
   LCS = 3.6000 + 1.9273(1.0900/6.9282**0.5) = 3.6000 + 0.3032 = 3.9032
   LCI = 3.6000 - 1.9273(1.0900/6.9282**0.5) = 3.6000 - 0.3032 = 3.2968

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1130$
   $\mu = 500 - 8(-1.2107) = 509.6858$

Nombre: ___ Valeria Higuera Manjarrez ___ Matrícula:___ A01742336 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(50.0000 < X < 50.0000) = P(1.0693 < Z < 1.0693) = 0.0000$

3. $X \sim Bin(n = 16, p = 0.05)$
   $P(X = 5) = 0.000776$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.6000-49.4000)/(6*0.1702) = 1.1750
   Cpks = (50.6000-49.8180)/(3*0.1702) = 1.5314
   Cpki = (49.8180-49.4000)/(3*0.1702) = 0.8186
   Cpk = min(1.5314, 0.8186) = 0.8186

5. **** Diferencia de medias
   $\nu = 80.5654$
   LCS = 3.60 - 3.0000 + 2.3739(0.2848) = 0.6000 + 0.6761 = 1.2761
   LCI = 3.60 - 3.0000 - 2.3739(0.2848) = 0.6000 - 0.6761 = -0.0761
   ** Antes del cambio
   LCS = 3.6000 + 2.4102(1.0900/6.8557**0.5) = 3.6000 + 0.3832 = 3.9832
   LCI = 3.6000 - 2.4102(1.0900/6.8557**0.5) = 3.6000 - 0.3832 = 3.2168

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1420$
   $\mu = 500 - 8(-1.0714) = 508.5710$

Nombre: ___ Katherine Ibarra Espinoza ___ Matrícula:___ A01742244 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.7000 < X < 50.4000) = P(-0.6932 < Z < 3.4193) = 0.7556$

3. $X \sim Bin(n = 17, p = 0.05)$
   $P(X = 6) = 0.000110$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.6000-49.4000)/(6*0.1702) = 1.1750
   Cpks = (50.6000-49.8180)/(3*0.1702) = 1.5314
   Cpki = (49.8180-49.4000)/(3*0.1702) = 0.8186
   Cpk = min(1.5314, 0.8186) = 0.8186

5. **** Diferencia de medias
   $\nu = 77.0626$
   LCS = 3.60 - 3.5300 + 1.9913(0.2911) = 0.0700 + 0.5796 = 0.6496
   LCI = 3.60 - 3.5300 - 1.9913(0.2911) = 0.0700 - 0.5796 = -0.5096
   ** Antes del cambio
   LCS = 3.6000 + 2.0154(1.0900/6.7082**0.5) = 3.6000 + 0.3275 = 3.9275
   LCI = 3.6000 - 2.0154(1.0900/6.7082**0.5) = 3.6000 - 0.3275 = 3.2725

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0440$
   $\mu = 500 - 8(-1.7060) = 513.6483$

Nombre: ___ Camila María Gámez Ramírez ___ Matrícula:___ A01742082 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.9000 < X < 50.4000) = P(0.4818 < Z < 3.4193) = 0.3147$

3. $X \sim Bin(n = 15, p = 0.05)$
   $P(X = 4) = 0.004853$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227

5. **** Diferencia de medias
   $\nu = 66.5541$
   LCS = 3.60 - 2.8800 + 1.6683(0.3127) = 0.7200 + 0.5216 = 1.2416
   LCI = 3.60 - 2.8800 - 1.6683(0.3127) = 0.7200 - 0.5216 = 0.1984
   ** Antes del cambio
   LCS = 3.6000 + 1.6860(1.0900/6.2450**0.5) = 3.6000 + 0.2943 = 3.8943
   LCI = 3.6000 - 1.6860(1.0900/6.2450**0.5) = 3.6000 - 0.2943 = 3.3057

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1590$
   $\mu = 500 - 8(-0.9986) = 507.9886$

Nombre: ___ Rodolfo Agustín Gámez Ramírez ___ Matrícula:___ A01742071 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.7000 < X < 50.1000) = P(-0.6932 < Z < 1.6568) = 0.7071$
3. $X \sim Bin(n = 15, p = 0.05)$
   $P(X = 4) = 0.004853$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227
5. **** Diferencia de medias
   $\nu = 77.0626$
   LCS = 3.60 - 3.4500 + 1.7741(0.2911) = 0.1500 + 0.5164 = 0.6664
   LCI = 3.60 - 3.4500 - 1.7741(0.2911) = 0.1500 - 0.5164 = -0.3664
   ** Antes del cambio
   LCS = 3.6000 + 1.7921(1.0900/6.7082**0.5) = 3.6000 + 0.2912 = 3.8912
   LCI = 3.6000 - 1.7921(1.0900/6.7082**0.5) = 3.6000 - 0.2912 = 3.3088
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1570$
   $\mu = 500 - 8(-1.0069) = 508.0549$

Nombre: ___ Celeste Carmona Ramírez ___ Matrícula:___ A01742044 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.6000 < X < 50.4000) = P(-1.2807 < Z < 3.4193) = 0.8995$

3. $X \sim Bin(n = 13, p = 0.05)$
   $P(X = 6) = 0.000019$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.6000-49.4000)/(6*0.1702) = 1.1750
   Cpks = (50.6000-49.8180)/(3*0.1702) = 1.5314
   Cpki = (49.8180-49.4000)/(3*0.1702) = 0.8186
   Cpk = min(1.5314, 0.8186) = 0.8186

5. **** Diferencia de medias
   $\nu = 71.8083$
   LCS = 3.60 - 2.9300 + 2.2146(0.3013) = 0.6700 + 0.6672 = 1.3372
   LCI = 3.60 - 2.9300 - 2.2146(0.3013) = 0.6700 - 0.6672 = 0.0028
   ** Antes del cambio
   LCS = 3.6000 + 2.2482(1.0900/6.4807**0.5) = 3.6000 + 0.3781 = 3.9781
   LCI = 3.6000 - 2.2482(1.0900/6.4807**0.5) = 3.6000 - 0.3781 = 3.2219

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1390$
   $\mu = 500 - 8(-1.0848) = 508.6786$

Nombre: ___ Valeria Montoya Valenzuela ___ Matrícula:___ A01743629 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(50.0000 < X < 50.0000) = P(1.0693 < Z < 1.0693) = 0.0000$
3. $X \sim Bin(n = 14, p = 0.05)$
   $P(X = 6) = 0.000031$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227
5. **** Diferencia de medias
   $\nu = 71.8083$
   LCS = 3.60 - 2.9600 + 1.9113(0.3013) = 0.6400 + 0.5759 = 1.2159
   LCI = 3.60 - 2.9600 - 1.9113(0.3013) = 0.6400 - 0.5759 = 0.0641
   ** Antes del cambio
   LCS = 3.6000 + 1.9343(1.0900/6.4807**0.5) = 3.6000 + 0.3253 = 3.9253
   LCI = 3.6000 - 1.9343(1.0900/6.4807**0.5) = 3.6000 - 0.3253 = 3.2747
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0450$
   $\mu = 500 - 8(-1.6954) = 513.5632$

Nombre: ___ Alejandro Castro Valenzuela ___ Matrícula:___ A01742274 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.7000 < X < 50.1000) = P(-0.6932 < Z < 1.6568) = 0.7071$

3. $X \sim Bin(n = 14, p = 0.05)$
   $P(X = 6) = 0.000031$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227

5. **** Diferencia de medias
   $\nu = 77.0626$
   LCS = 3.60 - 2.7900 + 1.9089(0.2911) = 0.8100 + 0.5556 = 1.3656
   LCI = 3.60 - 2.7900 - 1.9089(0.2911) = 0.8100 - 0.5556 = 0.2544
   ** Antes del cambio
   LCS = 3.6000 + 1.9305(1.0900/6.7082**0.5) = 3.6000 + 0.3137 = 3.9137
   LCI = 3.6000 - 1.9305(1.0900/6.7082**0.5) = 3.6000 - 0.3137 = 3.2863

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1580$
   $\mu = 500 - 8(-1.0027) = 508.0217$

Nombre: ___ María José López Cháidez ___ Matrícula:___ A01742039 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.6000 < X < 50.2000) = P(-1.2807 < Z < 2.2443) = 0.8875$

3. $X \sim Bin(n = 16, p = 0.05)$
   $P(X = 5) = 0.000776$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227

5. **** Diferencia de medias
   $\nu = 77.0626$
   LCS = 3.60 - 2.8000 + 1.7741(0.2911) = 0.8000 + 0.5164 = 1.3164
   LCI = 3.60 - 2.8000 - 1.7741(0.2911) = 0.8000 - 0.5164 = 0.2836
   ** Antes del cambio
   LCS = 3.6000 + 1.7921(1.0900/6.7082**0.5) = 3.6000 + 0.2912 = 3.8912
   LCI = 3.6000 - 1.7921(1.0900/6.7082**0.5) = 3.6000 - 0.2912 = 3.3088

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0810$
   $\mu = 500 - 8(-1.3984) = 511.1870$

Nombre: ___ Mia Teresa Beltrán Inzunza ___ Matrícula:___ A01743791 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.9000 < X < 50.0000) = P(0.4818 < Z < 1.0693) = 0.1725$

3. $X \sim Bin(n = 13, p = 0.05)$
   $P(X = 6) = 0.000019$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227

5. **** Diferencia de medias
   $\nu = 77.0626$
   LCS = 3.60 - 3.1300 + 1.8375(0.2911) = 0.4700 + 0.5348 = 1.0048
   LCI = 3.60 - 3.1300 - 1.8375(0.2911) = 0.4700 - 0.5348 = -0.0648
   ** Antes del cambio
   LCS = 3.6000 + 1.8571(1.0900/6.7082**0.5) = 3.6000 + 0.3018 = 3.9018
   LCI = 3.6000 - 1.8571(1.0900/6.7082**0.5) = 3.6000 - 0.3018 = 3.2982

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0420$
   $\mu = 500 - 8(-1.7279) = 513.8235$

Nombre: ___ Héctor Sepúlveda Báez ___ Matrícula:___ A01743909 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.6000 < X < 50.3000) = P(-1.2807 < Z < 2.8317) = 0.8975$

3. $X \sim Bin(n = 13, p = 0.05)$
   $P(X = 4) = 0.002816$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227

5. **** Diferencia de medias
   $\nu = 85.8197$
   LCS = 3.60 - 3.0200 + 2.3710(0.2761) = 0.5800 + 0.6547 = 1.2347
   LCI = 3.60 - 3.0200 - 2.3710(0.2761) = 0.5800 - 0.6547 = -0.0747
   ** Antes del cambio
   LCS = 3.6000 + 2.4049(1.0900/7.0711**0.5) = 3.6000 + 0.3707 = 3.9707
   LCI = 3.6000 - 2.4049(1.0900/7.0711**0.5) = 3.6000 - 0.3707 = 3.2293

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1790$
   $\mu = 500 - 8(-0.9192) = 507.3535$

Nombre: ___ José Antonio Salinas Cota ___ Matrícula:___ A01743555 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.6000 < X < 50.0000) = P(-1.2807 < Z < 1.0693) = 0.7574$
3. $X \sim Bin(n = 16, p = 0.05)$
   $P(X = 7) = 0.000006$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.6000-49.4000)/(6*0.1702) = 1.1750
   Cpks = (50.6000-49.8180)/(3*0.1702) = 1.5314
   Cpki = (49.8180-49.4000)/(3*0.1702) = 0.8186
   Cpk = min(1.5314, 0.8186) = 0.8186
5. **** Diferencia de medias
   $\nu = 78.8140$
   LCS = 3.60 - 3.2900 + 1.8371(0.2879) = 0.3100 + 0.5289 = 0.8389
   LCI = 3.60 - 3.2900 - 1.8371(0.2879) = 0.3100 - 0.5289 = -0.2189
   ** Antes del cambio
   LCS = 3.6000 + 1.8561(1.0900/6.7823**0.5) = 3.6000 + 0.2983 = 3.8983
   LCI = 3.6000 - 1.8561(1.0900/6.7823**0.5) = 3.6000 - 0.2983 = 3.3017
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1390$
   $\mu = 500 - 8(-1.0848) = 508.6786$

Nombre: ___ Santiago Peña Meza ___ Matrícula:___ A01742252 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.8000 < X < 50.4000) = P(-0.1058 < Z < 3.4193) = 0.5418$

3. $X \sim Bin(n = 15, p = 0.05)$
   $P(X = 6) = 0.000049$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.4000-49.6000)/(6*0.1702) = 0.7833
   Cpks = (50.4000-49.8180)/(3*0.1702) = 1.1398
   Cpki = (49.8180-49.6000)/(3*0.1702) = 0.4269
   Cpk = min(1.1398, 0.4269) = 0.4269

5. **** Diferencia de medias
   $\nu = 66.5541$
   LCS = 3.60 - 2.7500 + 1.9966(0.3127) = 0.8500 + 0.6242 = 1.4742
   LCI = 3.60 - 2.7500 - 1.9966(0.3127) = 0.8500 - 0.6242 = 0.2258
   ** Antes del cambio
   LCS = 3.6000 + 2.0244(1.0900/6.2450**0.5) = 3.6000 + 0.3533 = 3.9533
   LCI = 3.6000 - 2.0244(1.0900/6.2450**0.5) = 3.6000 - 0.3533 = 3.2467

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0800$
   $\mu = 500 - 8(-1.4051) = 511.2406$

Nombre: ___ Diego de Jesús Romero López ___ Matrícula:___ A01743796 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.7000 < X < 50.0000) = P(-0.6932 < Z < 1.0693) = 0.6134$
3. $X \sim Bin(n = 15, p = 0.05)$
   $P(X = 6) = 0.000049$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.6000-49.4000)/(6*0.1702) = 1.1750
   Cpks = (50.6000-49.8180)/(3*0.1702) = 1.5314
   Cpki = (49.8180-49.4000)/(3*0.1702) = 0.8186
   Cpk = min(1.5314, 0.8186) = 0.8186
5. **** Diferencia de medias
   $\nu = 64.8026$
   LCS = 3.60 - 3.5300 + 2.2195(0.3167) = 0.0700 + 0.7030 = 0.7730
   LCI = 3.60 - 3.5300 - 2.2195(0.3167) = 0.0700 - 0.7030 = -0.6330
   ** Antes del cambio
   LCS = 3.6000 + 2.2570(1.0900/6.1644**0.5) = 3.6000 + 0.3991 = 3.9991
   LCI = 3.6000 - 2.2570(1.0900/6.1644**0.5) = 3.6000 - 0.3991 = 3.2009
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1020$
   $\mu = 500 - 8(-1.2702) = 510.1619$

Nombre: ___ Manuel Brito Bon ___ Matrícula:___ A01741632 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.9000 < X < 50.2000) = P(0.4818 < Z < 2.2443) = 0.3026$

3. $X \sim Bin(n = 16, p = 0.05)$
   $P(X = 4) = 0.006147$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.7000-49.3000)/(6*0.1702) = 1.3708
   Cpks = (50.7000-49.8180)/(3*0.1702) = 1.7273
   Cpki = (49.8180-49.3000)/(3*0.1702) = 1.0144
   Cpk = min(1.7273, 1.0144) = 1.0144

5. **** Diferencia de medias
   $\nu = 61.2998$
   LCS = 3.60 - 3.2800 + 1.7803(0.3254) = 0.3200 + 0.5794 = 0.8994
   LCI = 3.60 - 3.2800 - 1.7803(0.3254) = 0.3200 - 0.5794 = -0.2594
   ** Antes del cambio
   LCS = 3.6000 + 1.8030(1.0900/6.0000**0.5) = 3.6000 + 0.3275 = 3.9275
   LCI = 3.6000 - 1.8030(1.0900/6.0000**0.5) = 3.6000 - 0.3275 = 3.2725

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1590$
   $\mu = 500 - 8(-0.9986) = 507.9886$

Nombre: ___ Miguel Ángel Figueroa Sánchez ___ Matrícula:___ A01743674 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.7000 < X < 50.2000) = P(-0.6932 < Z < 2.2443) = 0.7435$

3. $X \sim Bin(n = 16, p = 0.05)$
   $P(X = 7) = 0.000006$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.4000-49.6000)/(6*0.1702) = 0.7833
   Cpks = (50.4000-49.8180)/(3*0.1702) = 1.1398
   Cpki = (49.8180-49.6000)/(3*0.1702) = 0.4269
   Cpk = min(1.1398, 0.4269) = 0.4269

5. **** Diferencia de medias
   $\nu = 59.5484$
   LCS = 3.60 - 3.2100 + 2.3912(0.3300) = 0.3900 + 0.7892 = 1.1792
   LCI = 3.60 - 3.2100 - 2.3912(0.3300) = 0.3900 - 0.7892 = -0.3992
   ** Antes del cambio
   LCS = 3.6000 + 2.4411(1.0900/5.9161**0.5) = 3.6000 + 0.4498 = 4.0498
   LCI = 3.6000 - 2.4411(1.0900/5.9161**0.5) = 3.6000 - 0.4498 = 3.1502

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0420$
   $\mu = 500 - 8(-1.7279) = 513.8235$

Nombre: ___ Ricardo Blanco Tolosa ___ Matrícula:___ A00841820 ___

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
   LCS = 2.4933 + 3(0.0653/5**0.5) = 2.4933 + 0.0876 = 2.5809
   LCI = 2.4933 - 3(0.0653/5**0.5) = 2.4933 - 0.0876 = 2.4057

2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   $P(49.9000 < X < 50.3000) = P(0.4818 < Z < 2.8317) = 0.3127$

3. $X \sim Bin(n = 16, p = 0.05)$
   $P(X = 4) = 0.006147$

4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
   Cp = (50.5000-49.5000)/(6*0.1702) = 0.9792
   Cpks = (50.5000-49.8180)/(3*0.1702) = 1.3356
   Cpki = (49.8180-49.5000)/(3*0.1702) = 0.6227
   Cpk = min(1.3356, 0.6227) = 0.6227

5. **** Diferencia de medias
   $\nu = 71.8083$
   LCS = 3.60 - 2.6400 + 1.9113(0.3013) = 0.9600 + 0.5759 = 1.5359
   LCI = 3.60 - 2.6400 - 1.9113(0.3013) = 0.9600 - 0.5759 = 0.3841
   ** Antes del cambio
   LCS = 3.6000 + 1.9343(1.0900/6.4807**0.5) = 3.6000 + 0.3253 = 3.9253
   LCI = 3.6000 - 1.9343(1.0900/6.4807**0.5) = 3.6000 - 0.3253 = 3.2747

6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
   LCS = 0.0509 + 3((1-0.9213**2)**0.5)0.0552 = 0.0509 + 0.0645 = 0.1154
   LCS = 0.0509 - 3((1-0.9213**2)**0.5)0.0552 = 0.0509 - 0.0645 = -0.0136

7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1980$
   $\mu = 500 - 8(-0.8488) = 506.7903$